

Lab 02: Wireless Access Point

Configuring Wi-Fi Connectivity in Packet Tracer

Lab #2

Topic: Wireless
Networking

Tool: Cisco Packet Tracer

Cert: CompTIA Network+

1. Lab Objectives

- Add and configure a Wireless Access Point (WAP) in Packet Tracer.
- Connect wireless client devices (laptops) to the WAP using SSID and WPA2 security.
- Verify wireless clients can communicate with the wired LAN.
- Understand 802.11 wireless standards and Wi-Fi security protocols.
- Relate to Network+ objective: 2.4 – Given a scenario, install and configure wireless technologies.

2. Network Topology

A Wireless Access Point is connected via its Ethernet port to a Layer 2 switch. Two laptops connect wirelessly to the WAP. The switch also connects to a router providing the default gateway. All devices share the 192.168.10.0/24 subnet.

Device	Type	IP Address	Connection Method
Router0	Router (1841)	192.168.10.1	Wired – to SW1 Fa0/1
SW1	Switch (2960)	—	Wired backbone
AP0	Access Point	192.168.10.2	Wired – to SW1 Fa0/2
Laptop0	Wireless Client	192.168.10.10	Wi-Fi – SSID: LabWiFi
Laptop1	Wireless Client	192.168.10.11	Wi-Fi – SSID: LabWiFi

3. Step-by-Step Configuration

3.1 Access Point Setup

Click the Access Point → Config tab and configure:

SSID	LabWiFi
Authentication	WPA2-PSK
PSK Pass Phrase	Cisco123!

Encryption Type	AES
AP IP Address	192.168.10.2 / 255.255.255.0

3.2 Laptop Wireless NIC Configuration

For each laptop: Physical tab → remove Ethernet card → insert Wireless NIC (WPC300N). Then Desktop → PC Wireless → Connect tab:

1. Refresh network list – LabWiFi should appear
2. Select LabWiFi → Connect
3. Enter PSK: Cisco123!
4. Go to Desktop → IP Configuration:

Laptop0: 192.168.10.10 / 255.255.255.0 / GW 192.168.10.1

Laptop1: 192.168.10.11 / 255.255.255.0 / GW 192.168.10.1

3.3 Router Configuration

```
Router> enable
Router# configure terminal
Router(config)# interface FastEthernet0/0
Router(config-if)# ip address 192.168.10.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# exit
Router# write memory
```

4. Verification

```
Laptop0> ping 192.168.10.11    ! Laptop0 → Laptop1 (wireless to wireless)
Laptop0> ping 192.168.10.1    ! Laptop0 → Router  (wireless to wired)
```

5. Key Takeaways

- An Access Point bridges wireless clients onto the wired LAN — it operates at Layer 2.
- WPA2 with AES is the current industry standard for Wi-Fi security (WEP and TKIP are deprecated).
- SSID is the human-readable network name; it does not need to be broadcast for security but clients must know it.
- In Packet Tracer, laptops need a wireless NIC module installed before they can connect to a WAP.
- Always assign a strong PSK and avoid default SSID names in production environments.